

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2025

SP5 - Investment and Finance

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) Recently regulator has changed regulation and have introduced Bond forwards instead of FRA (Forward Rate Agreement). In just concluded ALCO (ALM Committee meeting), there was a discussion on the said topic. Your team wanted to understand the discussion and have asked the following 2 questions to you.

i) What is a Bond forward? (2)

ii) What is the difference between Bond forwards and FRAs? (6)

During the discussion on FRA portfolio, CRO has highlighted liquidity risk related to FRA portfolio and the importance of managing it.

iii) What are the liquidity risks in FRA portfolio and how can company mitigate them? (6)

[14 Marks]

Q.2) Recent tariff war has led to a lot of volatility across the global economies and have brought the focus back on Central Banks to help their government navigate the current situation. In India, RBI play an active role in monitoring and management of the Indian economy.

i) List the role of a Central Bank? (3)

ii) State how Central banks use non market (direct) controls? (5)

In the recently concluded MPC (Monetary Policy Committee) meeting of RBI, its minutes of the meeting captured the below statement:

“ After assessing the current and evolving macroeconomic situation, the MPC unanimously voted to reduce the policy repo rate by 25 basis points to 6.00 per cent with immediate effect. Consequently, the standing deposit facility (SDF) rate under the liquidity adjustment facility (LAF) shall stand adjusted to 5.75 per cent and the marginal standing facility (MSF) rate and the Bank Rate to 6.25 per cent. This decision is in consonance with the objective of achieving the medium-term target for consumer price index (CPI) inflation of 4 per cent within a band of +/- 2 per cent, while supporting growth.”

iii) Discuss the impact of RBI's decision? (9)

iv) You are an Investment Actuary of a Life Insurance Co.. CRO has asked you for a short note on the areas that the company should be concerned about due to RBI's decision. What all aspects would you cover in your note? (9)

[26 Marks]

Q.3) It's the year end and CHRO is busy with performance assessment exercise. CIO has given the rating and promotion list to CHRO. Two fund managers have been given the top rating and both have been proposed for promotion. CHRO is not convinced and wants to differentiate between the fund managers basis their investment performance. You are the CRO of the company and CHRO has come to you for help. You have suggested CHRO to look at Information ratio (IR) of the funds of both the managers for make a decision in respect to which of the fund manager has performed well and can be given a promotion.

i) What is IR ? (2)

ii) Why would have CRO suggested IR for assessment and why it may not be a good measure? (8)

iii) Following information is available for both the fund managers. Calculate and comment on their relative performance? (8)

Year	Benchmark Return (%)	Fund X Return (%)	Fund Y Return (%)
1	10%	12%	9%
2	8%	9%	7%
3	5%	8%	5%
4	7%	6%	8%

[18 Marks]

Q.4) To what extent does the structure and incidence of taxation affect investment decision-making processes, and what approaches can be employed to assess the influence of tax policies on investment behaviour? [4 Marks]

Q.5) Amid heightened turmoil in domestic equities, a wealth manager to a wealthy US investor has suggested that portfolio needs some international diversification.

- i) To what extent can international diversification improve the portfolio's risk-return profile, and what factors should the manager critically evaluate before implementation? (8)

The client suggested taking exposure through following types of derivatives:

- Using futures market
- Using Options
- Undertaking an Equity swaps

- ii) Describe how the manager might achieve international exposure through each of the three routes suggested above? (5)

[13 Marks]

Q.6) You are an Investment Actuary in one of the country's largest life insurance companies, which sells a host of products like Protection, Health, Traditional Savings product and Unit Linked products. During Investment Committee discussion, one of the Independent directors commented that commodities should be looked as an investment class. Your boss has asked for a discussion on the comment.

- i) What is a Commodity? (2)
- ii) What is meant by Institutional Investment in commodities and what are the ways in which they can invest in commodities? (4)
- iii) Your boss is exploring use of Commodities as an asset class to enhance the returns. He also suggested that higher returns will help reduce capital requirement for the company thereby able to demonstrate better financial strength. Discuss his comments. He has suggested the investment as an Institutional investor. (14)
- iv) A big Indian brokerage house has famously said that they have never raised Venture capital money. Discuss the reasons as to why they went for such an approach? (5)

[25 Marks]
